

WSB UNIVERSITY · TEAM PROJECT — SEMESTER 4

Data Mining Pipeline

Technical Report — End-to-End Implementation

PREPARED BY

Güner Özben · 65281

Aras Taşdemir · 65856

Berkay Paray · 62403

SCOPE

Five data-mining sub-tasks

Collection · Preprocessing · Feature engineering · Visualization



From Raw Data to Actionable Insight

A data mining pipeline is a structured, end-to-end workflow that transforms raw, unprocessed data into meaningful insights and actionable knowledge — a sequence of well-defined stages guiding the analytical process from problem definition to decision-making.



Data Mining (KDD)

Extracting hidden patterns, correlations, and algorithmic relationships within large datasets.



Data Science Pipeline

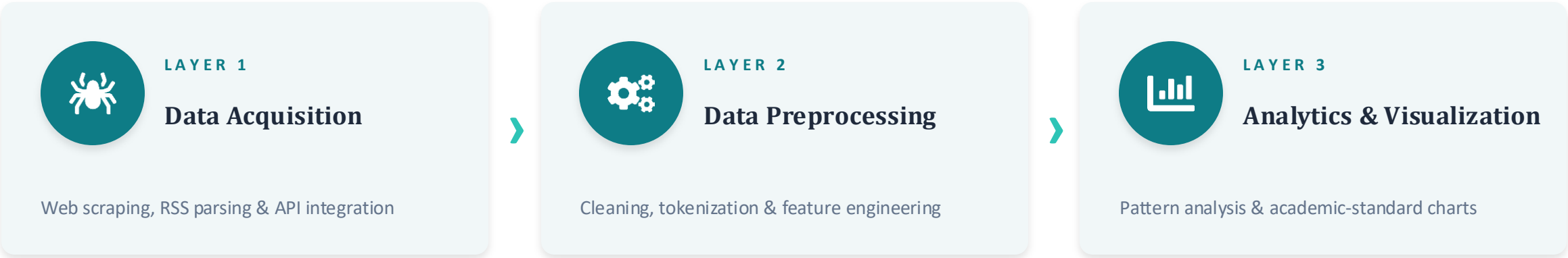
The full lifecycle: problem definition, data engineering, modeling, deployment, and monitoring.



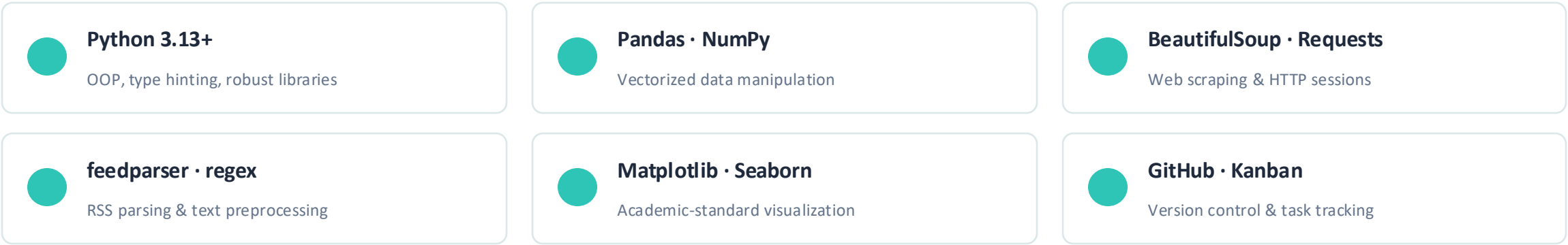
Report Scope

A detailed technical overview of five sub-tasks — Web Scraping of Job Listings, News Headlines Analyzer, Product Price Tracker, Weather Data Collection, and a COVID-19 Public Health Dashboard — covering data collection, preprocessing, feature engineering, and pattern visualization.

Three-Layer Pipeline & Technology Stack



Technology Stack



Web Scraping of Job Listings

```
import requests, pandas as pd
from bs4 import BeautifulSoup

def scrape_jobs(url, pages=5):
    rows = []
    for p in range(1, pages + 1):
        html = requests.get(f"{url}?page={p}").text
        soup = BeautifulSoup(html, "html.parser")
        for c in soup.select(".job-card"):
            rows.append({"title": c.select_one(".title").text,
                        "skills": c.select_one(".skills").text})
    return pd.DataFrame(rows)

df = scrape_jobs("https://jobs.example.com")
skills = df["skills"].str.split(",").explode().str.strip()
top_skills = skills.value_counts().head(8)
```



MOST IN-DEMAND TECH SKILLS



Insight Python, SQL and cloud platforms stand out as the dominant skills across modern tech positions.

News Headlines Analyzer



headlines.py

```
import re, feedparser
from collections import Counter

STOP = {"the", "a", "to", "of", "in", "on", "and", "for"}

def clean(text):
    text = re.sub(r"[^a-z0-9\s]", "", text.lower())
    return [w for w in text.split() if w not in STOP]

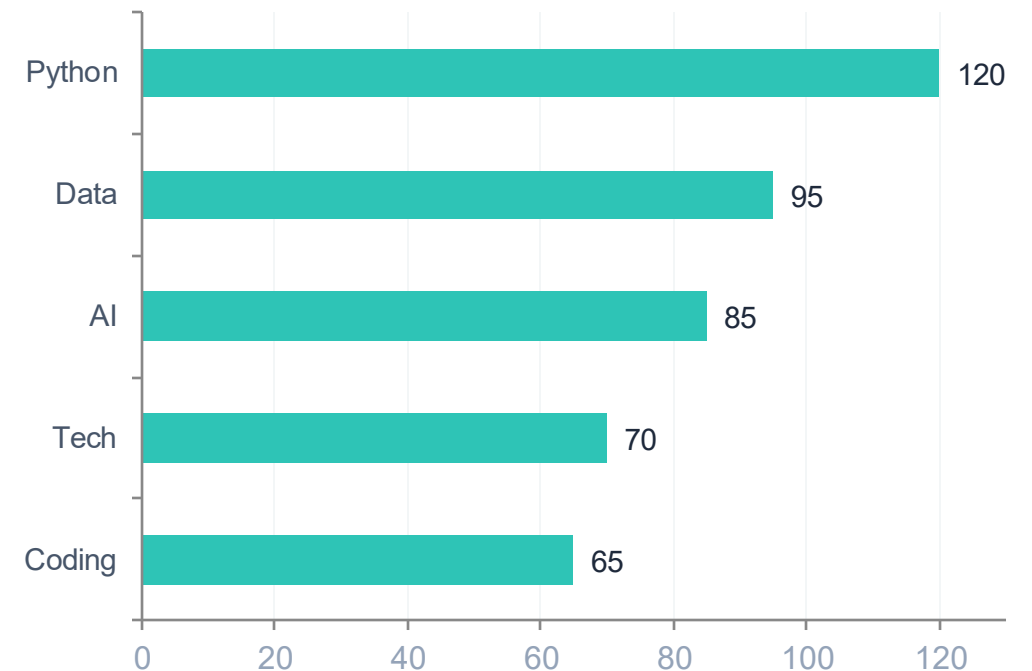
feeds = ["https://news.ycombinator.com/rss",
         "http://feeds.bbc.co.uk/news/rss.xml"]

words = []
for url in feeds:
    for entry in feedparser.parse(url).entries:
        words += clean(entry.title)

top10 = Counter(words).most_common(10)
```



TOP KEYWORDS ACROSS NEWS CHANNELS



Insight Tech-centric vocabulary — led by Python, Data and AI — dominates aggregated headlines.

Product Price Tracker



price_tiers.py

```
import pandas as pd

def to_tier(price):
    if price < 15: return "Low"
    if price <= 35: return "Medium"
    return "High"

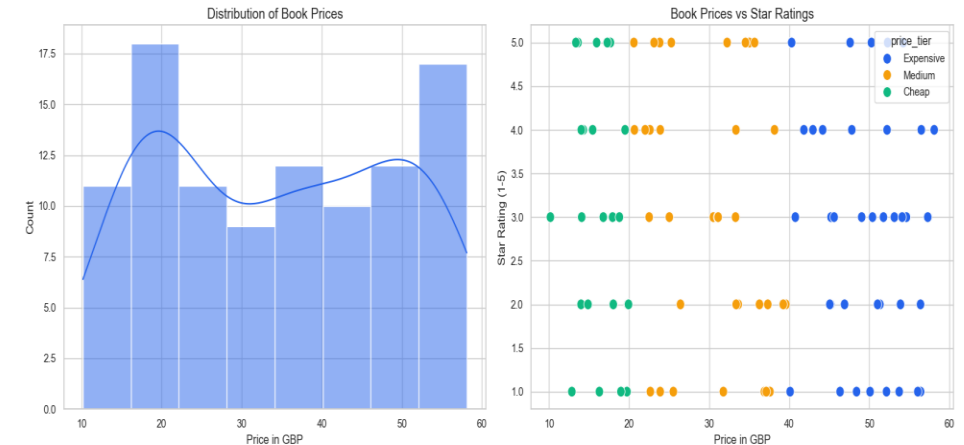
df = pd.DataFrame(scraped_products) # title, price, rating
df["price_tier"] = df["price"].apply(to_tier)

summary = (df.groupby("price_tier")
            .agg(avg_price=("price", "mean"),
                 avg_rating=("rating", "mean"),
                 products=("title", "size"))
            .round(2))
```



PRICE TIER ANALYSIS (BOOKS SAMPLE)

Price Tier	Avg Price	Avg Rating	Products
Low (< £15)	£10.45	3.12	34
Medium (£15–35)	£24.80	3.45	45
High (> £35)	£42.15	3.68	21



Insight Higher price tiers correlate with higher average star ratings across sampled products.

Weather Data Collection & Analysis



weather_client.py

```
import requests, pandas as pd

class WeatherAPIClient:
    BASE = "https://archive-api.open-meteo.com/v1/archive"

    def __init__(self):
        self.session = requests.Session()

    def fetch(self, lat, lon, start, end):
        params = {"latitude": lat, "longitude": lon,
                  "start_date": start, "end_date": end,
                  "daily": "temperature_2m_max,precipitation_sum"}
        r = self.session.get(self.BASE, params=params, timeout=30)
        r.raise_for_status()
        return pd.DataFrame(r.json()["daily"]).set_index("time")

df = WeatherAPIClient().fetch(50.06, 19.94,
                              "2026-06-01", "2026-06-08")
```



WEATHER CONDITION CATEGORIES

Condition	Avg Temp	Rainfall	Hours
Clear / Sunny	22.4 °C	0.0 mm	72 h
Partly Cloudy	17.1 °C	2.4 mm	56 h
Overcast & Rain	12.5 °C	34.8 mm	40 h

Insight Average temperature falls sharply as total accumulated rainfall rises across categories.

COVID-19 & Public Health Dashboard



covid_dashboard.py

```
import pandas as pd

URL = "https://covid.ourworldindata.org/data/owid-covid-data.csv"
df = pd.read_csv(URL)

latest = (df.sort_values("date")
          .groupby("location").last()
          .dropna(subset=["total_cases", "total_deaths"]))

latest["mortality_rate"] = (latest["total_deaths"]
                           / latest["total_cases"] * 100).round(2)

top = latest.sort_values("total_cases",
                        ascending=False).head(5)
```



TOP COUNTRIES BY TOTAL CASES

Country	Cases	Deaths	Mortality
United States	100.0M	1.10M	1.10%
India	45.0M	0.53M	1.18%
Brazil	37.0M	0.70M	1.89%
France	35.0M	0.16M	0.46%
Germany	32.0M	0.17M	0.53%

Insight Brazil shows the highest computed mortality rate among the highest-case countries.

Automated Data Pipeline Experiment

An independent Books Scraping Pipeline was integrated to test the automation and state-management capabilities of the data mining pipeline — extracting and cleaning unstructured web data, then producing output in two distinct formats.



Structured Storage

`scraped_books_data.csv`

Book name, price, rating & stock status as a database table



Metadata Registry

`books_pipeline_summary.json`

Operating statistics, log records & execution metrics



Pipeline Status: **SUCCESS**

`Books Scraper and Analyzer Pipeline · executed 2026-06-08`

1,000

Records Scraped

50

Categories Analyzed

3.54

Average Rating

12.45s

Processing Time

Academic Integrity & Collaboration



Fair Task Distribution

Members owned their modules while sharing joint responsibility for the consistency of the whole pipeline architecture.



Kanban Project Management

A GitHub Projects board tracked task state transitions across Todo, In Progress, and Done in real time.



Code Review & Citation

Review processes prevented code conflicts; all external datasets and libraries were fully referenced and cited.

CONCLUSION

A Reliable End-to-End Foundation

By implementing diverse components across different kinds of data — web scraping, RSS parsing, API connectivity, and public datasets — the project proves the effectiveness of systematic data pipelines. Structured preprocessing is vital to turn noisy, real-world data into clear, actionable business intelligence.

FUTURE WORK



Workflow Orchestration

Fully automate static pipelines with Apache Airflow



Machine Learning Layer

Add anomaly detection & forecasting via Prophet / Isolation Forest



Interactive Dashboards

Migrate to Streamlit or Dash for real-time web interaction